

参考资料:

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谢惠民, 恽自求, 钱定边, 易法槐.

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上册: 数学分析; 下册: 高等代数与空间解析几何. 科学出版社. 2024. (难度上来了!)

Recall: 空间曲线之切线与法平面

空间曲线 Γ 参数方程:

$$\vec{r} = \vec{r}(t) = \{x(t), y(t), z(t)\}, \quad t \in I \subset \mathbb{R}.$$

表示质点在空间运动, 在 t 时刻所处位置.

$$\begin{cases} x = x(t), \\ y = y(t), \\ z = z(t) \end{cases}$$

空间曲线 Γ 是质点运动之轨迹.

$$\vec{r}'(t) = \{x'(t), y'(t), z'(t)\} \neq \vec{0}. \quad \text{运动方向.}$$

—— 切线方向.

空间曲线 Γ 光滑 ^{定义} $\Leftrightarrow x(t), y(t), z(t)$ 连续可导,
且 $\vec{r}'(t) \neq 0$.

空间曲线 Γ 的切线方程:

在 t_0 处: $\underline{P_0(x_0, y_0, z_0) \in \Gamma}$:
$$\begin{cases} x_0 = x(t_0) \\ y_0 = y(t_0) \\ z_0 = z(t_0) \end{cases}$$

切线方向 $\underline{\vec{r}'(t_0)} = \{x'(t_0), y'(t_0), z'(t_0)\}$.

切线方程:
$$\frac{x-x_0}{x'(t_0)} = \frac{y-y_0}{y'(t_0)} = \frac{z-z_0}{z'(t_0)}.$$

法平面, ———— 过切点 P_0 且与切线垂直之平面.

法平面方程是.

$$x'(t_0)(x-x_0) + y'(t_0)(y-y_0) + z'(t_0)(z-z_0) = 0$$

如果曲线 Γ 是两个空间曲面之交线, 则可表示为:

$$\Gamma: \begin{cases} F_1(x, y, z) = 0 & \longleftarrow \text{曲面 } S_1 \\ F_2(x, y, z) = 0 & \longleftarrow \text{曲面 } S_2 \end{cases}$$

F_1, F_2 连续可微. $P_0(x_0, y_0, z_0) \in \Gamma$.

Jacobi 行列式 $J = \frac{\partial(F_1, F_2)}{\partial(y, z)} \Big|_{p_0} = \begin{vmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{vmatrix} \Big|_{p_0} \neq 0$

$\Rightarrow$ 存在函数组 $\begin{cases} y=y(x) \\ z=z(x) \end{cases} \quad x \in I, \text{ 可微. } \uparrow$
 (其存在性以后证明)

$$\begin{cases} F_1(x, y(x), z(x)) = 0 \\ F_2(x, y(x), z(x)) = 0 \end{cases} \quad x \in I.$$

两边对 x 求导数:

$$\begin{cases} F_{1x} + F_{1y} \cdot y'(x) + F_{1z} \cdot z'(x) = 0 \\ F_{2x} + F_{2y} \cdot y'(x) + F_{2z} \cdot z'(x) = 0 \end{cases}$$

$$J \neq 0 \Rightarrow \exists \quad y'(x) = \frac{1}{J} \begin{vmatrix} -F_{1x} & F_{1z} \\ -F_{2x} & F_{2z} \end{vmatrix} = \frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(z, x)}$$

$$z'(x) = \frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(x, y)}.$$

参数方程: $\vec{r}(x) = \{x, y(x), z(x)\},$

切线方向: $\vec{r}'(x) = \{1, y'(x), z'(x)\}$
 $= \left\{ 1, \frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(z, x)}, \frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(x, y)} \right\}$

或 $J \vec{r}'(x) = \left\{ \underbrace{\frac{\partial(F_1, F_2)}{\partial(y, z)}}_{\neq 0}, \frac{\partial(F_1, F_2)}{\partial(z, x)}, \frac{\partial(F_1, F_2)}{\partial(x, y)} \right\}$

$$(y, z), (z, x), (x, y)$$

$$x \quad y \quad z \quad x \quad y \quad z$$

在 $P_0(x_0, y_0, z_0) \in \Gamma$ 处, 满足 $\begin{cases} F_1(P_0) = 0 \\ F_2(P_0) = 0 \end{cases}$

切线方程:

$$J\vec{r}'(x)|_{P_0} = \left\{ \frac{\partial(F_1, F_2)}{\partial(y, z)}|_{P_0}, \frac{\partial(F_1, F_2)}{\partial(z, x)}|_{P_0}, \frac{\partial(F_1, F_2)}{\partial(x, y)}|_{P_0} \right\}$$

切线方程:

$$\frac{x - x_0}{\frac{\partial(F_1, F_2)}{\partial(y, z)}|_{P_0}} = \frac{y - y_0}{\frac{\partial(F_1, F_2)}{\partial(z, x)}|_{P_0}} = \frac{z - z_0}{\frac{\partial(F_1, F_2)}{\partial(x, y)}|_{P_0}}$$

$$\left(\left\{ \frac{\partial(F_1, F_2)}{\partial(y, z)}|_{P_0}, \frac{\partial(F_1, F_2)}{\partial(z, x)}|_{P_0}, \frac{\partial(F_1, F_2)}{\partial(x, y)}|_{P_0} \right\} \neq \vec{0} \right)$$

法平面方程:

$$\frac{\partial(F_1, F_2)}{\partial(y, z)}|_{P_0} (x - x_0) + \frac{\partial(F_1, F_2)}{\partial(z, x)}|_{P_0} (y - y_0) + \frac{\partial(F_1, F_2)}{\partial(x, y)}|_{P_0} (z - z_0) = 0$$

例 求曲线 $\Gamma: \begin{cases} x^2 + y^2 = a^2 \\ x^2 + z^2 = a^2 \end{cases}$ 在 $P_0(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}})$ 处

切线方程和法平面方程. ($a > 0$)

解: 首先: $P_0 \in \Gamma: \left(\frac{a}{\sqrt{2}}\right)^2 + \left(\frac{a}{\sqrt{2}}\right)^2 = a^2$

$$\text{令 } \begin{cases} F_1(x, y, z) = 0 : F_1 = x^2 + y^2 - a^2 = 0 \\ F_2(x, y, z) = 0 : F_2 = x^2 + z^2 - a^2 = 0 \end{cases}$$

$$\text{在 } P_0 \text{ 处: } \frac{\partial(F_1, F_2)}{\partial(y, z)} \Big|_{P_0} = \begin{vmatrix} 2y & 0 \\ 0 & 2z \end{vmatrix}_{y=z=\frac{a}{\sqrt{2}}} = 4yz = 2a^2 > 0$$

$$\frac{\partial(F_1, F_2)}{\partial(z, x)} \Big|_{P_0} = \begin{vmatrix} F_{1z} & F_{1x} \\ F_{2z} & F_{2x} \end{vmatrix}_{P_0} = \begin{vmatrix} 0 & 2x \\ 2z & 2x \end{vmatrix}_{x=z=\frac{a}{\sqrt{2}}} = -4xz = -2a^2 \neq 0$$

$$\frac{\partial(F_1, F_2)}{\partial(x, y)} \Big|_{P_0} = \begin{vmatrix} F_{1x} & F_{1y} \\ F_{2x} & F_{2y} \end{vmatrix}_{P_0} = \begin{vmatrix} 2x & 2y \\ 2x & 0 \end{vmatrix}_{x=y=\frac{a}{\sqrt{2}}} = -4xy = -2a^2$$

$$\therefore \text{切线方向: } \{2a^2, -2a^2, -2a^2\} = 2a^2 \{1, -1, -1\}$$

$$\text{切线方程为: } \frac{x - \frac{a}{\sqrt{2}}}{1} = \frac{y - \frac{a}{\sqrt{2}}}{-1} = \frac{z - \frac{a}{\sqrt{2}}}{-1}$$

$$\text{即 } x - \frac{a}{\sqrt{2}} = -(y - \frac{a}{\sqrt{2}}) = -(z - \frac{a}{\sqrt{2}})$$

$$\text{法平面: } 1(x - \frac{a}{\sqrt{2}}) + (-1)(y - \frac{a}{\sqrt{2}}) + (-1)(z - \frac{a}{\sqrt{2}}) = 0$$

$$\text{即: } x - y - z + \frac{a}{\sqrt{2}} = 0 \quad \#$$

§15.2. 曲面之切平面与法线

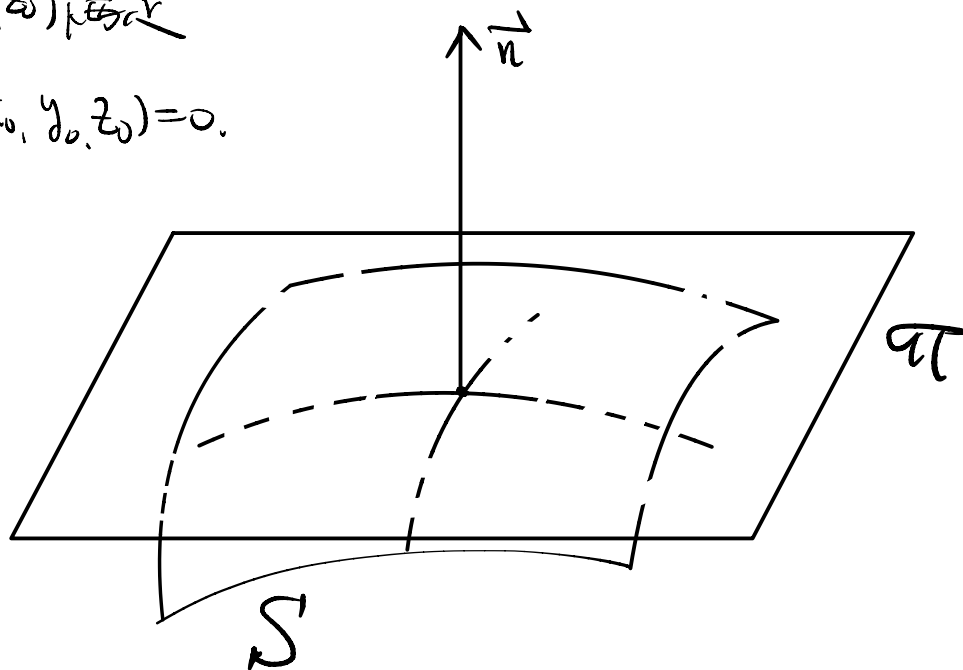
设曲面 S 的方程是:

$$F(x, y, z) = 0.$$

其中 $F(x, y, z)$ 有连续偏导数, 点 $P_0 \in S$,

即 $P_0(x_0, y_0, z_0)$ 满足

$$F(x_0, y_0, z_0) = 0.$$



若一个质点在 S 上运动, 过 $P_0(x_0, y_0, z_0)$ 点, 运动轨迹:

$\vec{r}(t) = \{x(t), y(t), z(t)\}$ 是光滑曲线 Γ , 运动方向

是: $\vec{r}'(t) = \{x'(t), y'(t), z'(t)\} \neq \vec{0}$. 对过 P_0 点,

即 $x_0 = x(t_0)$, $y_0 = y(t_0)$, $z_0 = z(t_0)$. $P(x(t), y(t), z(t))$

$P \in \Gamma \subset S$: 则 $x(t), y(t), z(t)$ 满足:

$$F(x(t), y(t), z(t)) = 0.$$

对 γ 有:

$$F_x \cdot x'(t) + F_y \cdot y'(t) + F_z \cdot z'(t) = 0.$$

在 $P_0(x_0, y_0, z_0)$:

$$F_x(P_0) \cdot x'(t_0) + F_y(P_0) \cdot y'(t_0) + F_z(P_0) \cdot z'(t_0) = 0.$$

$$\{F_x(P_0), F_y(P_0), F_z(P_0)\} \cdot \{x'(t_0), y'(t_0), z'(t_0)\} = 0$$

即 $\{F_x(P_0), F_y(P_0), F_z(P_0)\} \perp \{x'(t_0), y'(t_0), z'(t_0)\}$

若记 $\vec{n} \triangleq \{F_x(P_0), F_y(P_0), F_z(P_0)\} \neq 0$, 则对任何

在 S 上过 P_0 的任意运动, 其运动方向

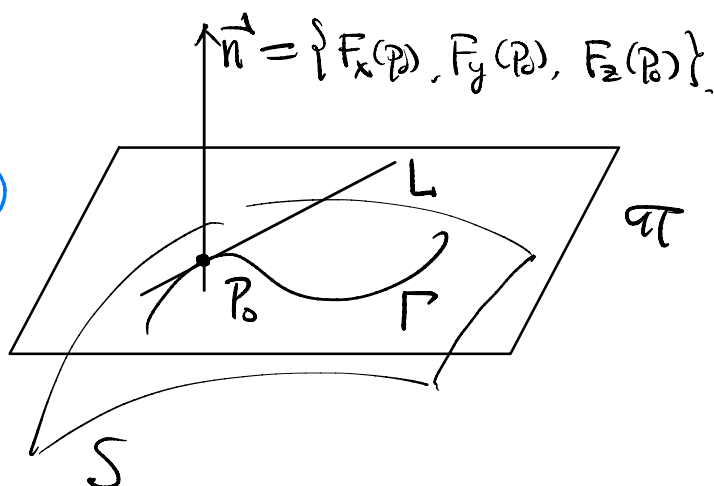
即曲线切线方向 $\vec{r}'(t_0) = \{x'(t_0), y'(t_0), z'(t_0)\}$

与 $\vec{n}$ 垂直:

$$\vec{n} \perp \vec{r}'(t_0).$$

任意光滑曲线之切方向

$\therefore$ 过 P_0 的任意光滑曲线之切线在同一个平面上:



过 $P_0(x_0, y_0, z_0)$ 点以 $\vec{n}$ 为法方向之平面,

称为 S 在 P_0 处之切平面.

切平面之法向: $\vec{n} = \{F_x(P_0), F_y(P_0), F_z(P_0)\} \neq 0$

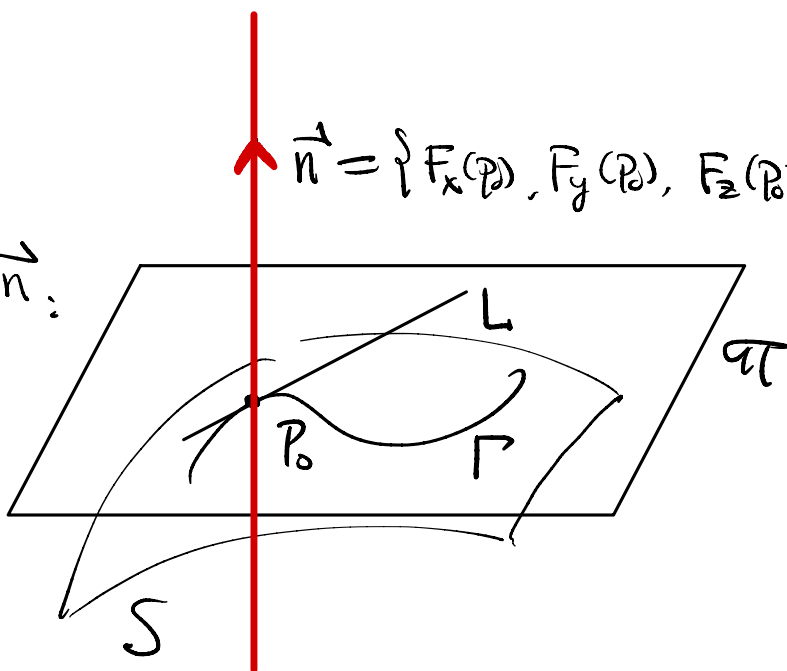
切平面方程:

$$F_x(P)(x-x_0) + F_y(P)(y-y_0) + F_z(P)(z-z_0) = 0.$$

过 $P_0(x_0, y_0, z_0)$ 点与切平面垂直的直线称之为 S 在 P_0 点

法线.

法线方向即 $\vec{n}$:



法线方程: $\frac{x-x_0}{F_x(P)} = \frac{y-y_0}{F_y(P)} = \frac{z-z_0}{F_z(P)}$

如果 S 是由函数

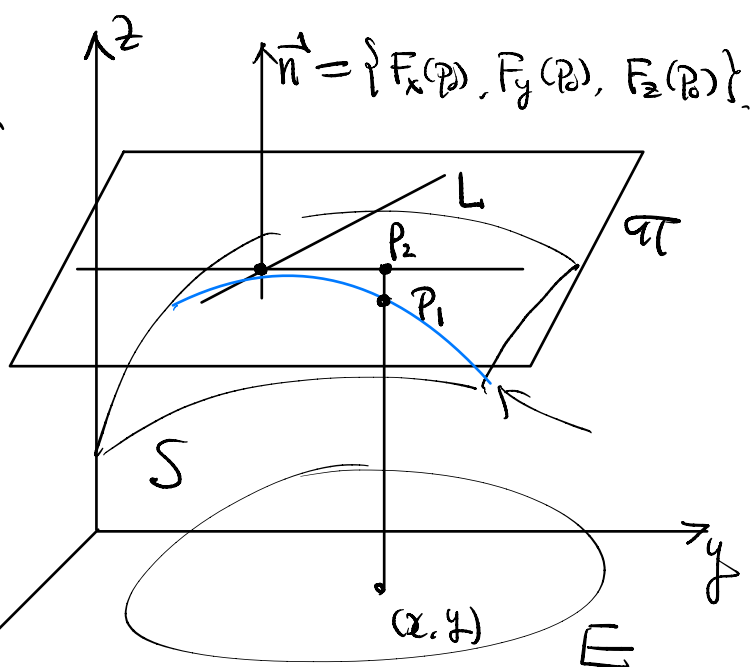
$$z = f(x, y), (x, y) \in E.$$

表示.

设

$$P_1(x, y, z = f(x, y)) \in S,$$

$$P_2(x, y, z) \in \pi:$$



$S: z = f(x, y)$, ———— 曲面:

$$f(x, y) - z = 0 \iff F(x, y, z) \triangleq f(x, y) - z$$

$$P_0(x_0, y_0, z_0) \in S: \quad z_0 = f(x_0, y_0).$$

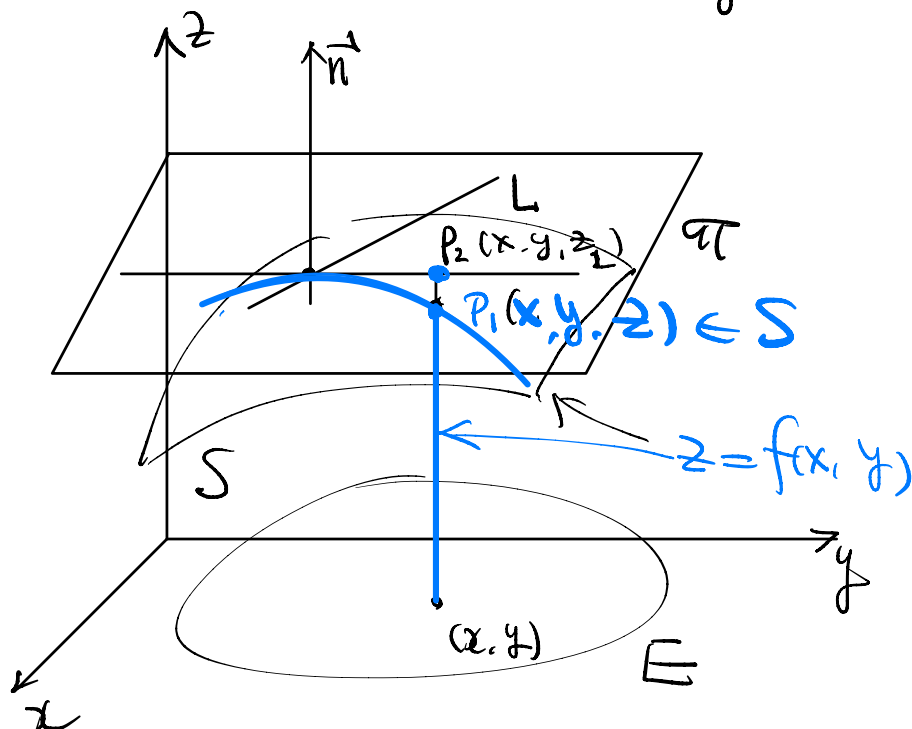
在 P_0 处平面法向量: $\vec{n} = \{F_x(P_0), F_y(P_0), F_z(P_0)\}$.

$$\vec{n} = \{f_x(x_0, y_0), f_y(x_0, y_0), -1\} \neq \vec{0}.$$

切平面方程:

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - \underbrace{(z - z_0)}_x = 0$$

$$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$



$P_2(x, y, z_2) \in \pi$: P_2 在切平面上: 则

$$z_2 - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$P_1(x, y, z) \in S: \quad z = f(x, y); \quad z_0 = f(x_0, y_0)$$

$$z - z_0 = f(x, y) - f(x_0, y_0)$$

$$= \underbrace{f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}_{p = \sqrt{(x-x_0)^2 + (y-y_0)^2}} + \underbrace{o(p)}_{\uparrow}$$

$$= z_2 - z_0 + o(p).$$

$$z = z_2 + o(p).$$

$\uparrow$
 $f(x, y)$ 切平面上.

$$z - z_2 = o(p). \quad (p \rightarrow 0).$$

$\therefore$ 切平面是曲面在点 P_0 的切线.

例. 求椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ ($a, b, c > 0$)

在 $P_0(x_0, y_0, z_0)$ 处的切平面方程.

解: $P_0(x_0, y_0, z_0)$ 在椭球面上: 则有

$$\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2} = 1.$$

$$\text{令 } F(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$$

$$\text{椭球面方程} \Leftrightarrow F(x, y, z) = 0.$$

切平面的法向: $\vec{n} = \{F_x(P_0), F_y(P_0), F_z(P_0)\} \neq \vec{0}.$

$$F_x = \frac{2x}{a^2}, \quad F_y = \frac{2y}{b^2}, \quad F_z = \frac{2z}{c^2},$$

$$\therefore \text{在 } P_0 \text{ 点: } \vec{n} = \left\{ \frac{2x_0}{a^2}, \frac{2y_0}{b^2}, \frac{2z_0}{c^2} \right\} \neq \vec{0}$$

$\therefore$ 切平面方程:

$$\frac{2x_0}{a^2}(x-x_0) + \frac{2y_0}{b^2}(y-y_0) + \frac{2z_0}{c^2}(z-z_0) = 0$$

$$\frac{x_0 x}{a^2} - \frac{x_0^2}{a^2} + \frac{y_0 y}{b^2} - \frac{y_0^2}{b^2} + \frac{z_0 z}{c^2} - \frac{z_0^2}{c^2} = 0$$

$\therefore$ 切平面方程:

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} + \frac{z_0 z}{c^2} = 1. \quad \#$$

§15.3. 方向导数和梯度.

考虑函数 $z = f(x, y)$, $(x, y) \in E$.

Recall 偏导数: 在 $(x_0, y_0) \in E$ 处 自变量增量:

$$(\Delta x, \Delta y) \triangleq (x - x_0, y - y_0):$$

函数增量

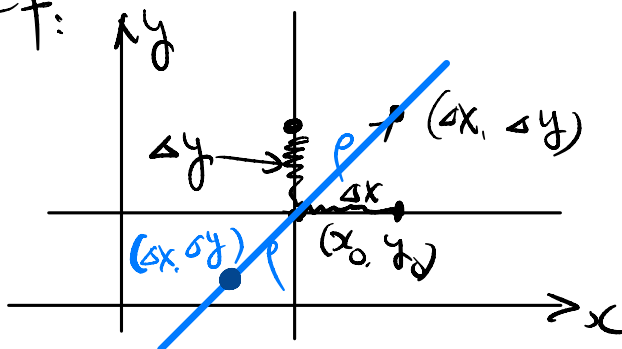
$$\Delta z = f(x, y) - f(x_0, y_0) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

若 $\Delta y = 0$: 函数关于 x 的变化率:

$$\frac{\Delta z}{\Delta x} \xrightarrow{\Delta x \rightarrow 0} \frac{\partial z}{\partial x};$$

$\Delta x = 0$:

$$\frac{\Delta z}{\Delta y} \xrightarrow{\Delta y \rightarrow 0} \frac{\partial z}{\partial y}.$$



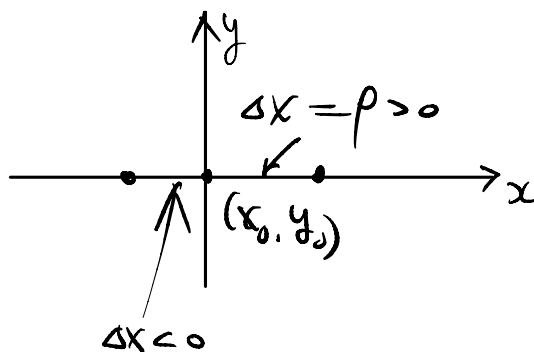
一般情况: 怎样观察变化率?

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

$$(\Delta x, \Delta y) \rightarrow (0, 0) \Leftrightarrow \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2} \rightarrow 0$$

$$\frac{\Delta z}{\rho} \xrightarrow{\rho \rightarrow 0} ?$$

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$



$$\frac{\Delta z}{\rho}$$

若 $\Delta y = 0$, $\rho = |\Delta x| \rightarrow 0$ 时左右极限:

$$\frac{\Delta z}{\rho} = \frac{\Delta z}{\Delta x} \xrightarrow{\Delta x \rightarrow 0^+} \frac{\partial z}{\partial x} \quad \checkmark$$

左极限: $\rho = \sqrt{(\Delta x)^2} = |\Delta x| = -\Delta x \rightarrow 0 \ (\Delta x \rightarrow 0^-)$:

$$\frac{\Delta z}{\rho} = \frac{\Delta z}{-\Delta x} \xrightarrow{\Delta x \rightarrow 0^-} -\frac{\partial z}{\partial x} \quad (\text{与右极限不一致!})$$

所以“变化率”有方向性:

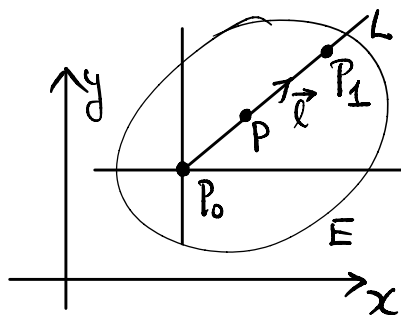
$$\left(\neq \frac{\partial z}{\partial x} \right)$$

定义 3.1 设 $z = f(x, y)$ 定义于 $E \subset \mathbb{R}^2$. $P_0(x_0, y_0) \in E$,
从 P_0 出发的射线 L , 其方向为:

$$\vec{l} = \{\cos \alpha, \cos \beta\} \quad (\alpha, \beta \text{ 是 } \vec{l} \text{ 的}$$

方向角, 即与 x 轴, y 轴的夹

角, $\cos \alpha, \cos \beta$ 是 $\vec{l}$ 的方向余弦)



$$L \text{ 的参数方程为: } \begin{cases} x = x(t) \triangleq x_0 + t \cos \alpha, \\ y = y(t) \triangleq y_0 + t \cos \beta, \end{cases}$$

$$P_1(x(t_1), y(t_1))$$

$$P_0 P_1 \subset E.$$

$0 \leq t \leq t_1$ 时, $(x(t), y(t)) \in L \cap E$. 如果取点

$$\lim_{t \rightarrow 0^+} \frac{f(x(t), y(t)) - f(x_0, y_0)}{t} = A,$$

存在, A 称为 $f(x, y)$ 在 (x_0, y_0) 沿方向 $\vec{l}$ 的方向导数.

记为: $\frac{\partial f}{\partial \vec{l}}|_{P_0}$, 即

$$\frac{\partial f}{\partial \vec{l}}|_{P_0} = \lim_{\substack{t \rightarrow 0 \\ P \in L \\ P \rightarrow P_0}} \frac{f(x(t), y(t)) - f(x_0, y_0)}{t} = \lim_{\substack{P \in L \\ P \rightarrow P_0}} \frac{f(P) - f(P_0)}{|P_0 P|}.$$

作业: P.129 2, 3. P132 1-4.